SIMATS ENGINEERING

CSA15 – CLOUD COMPUTING AND BIG DATA ANALYTICS

ASSIGNMENT 3 – CO3
OBE-Based Evidence Principle

Design a Cloud-Based Collaborative Solution
CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.*

Field	Details	Field	Details
Student Name	M.V. Raj Varun	Register Number	192411007
Department / Section	CSE	Academic Year / Semester	MS2 Sep-Dec 2026
Faculty Name	Nattar Kannan	Nature of Assignment	Individual
Duration	1 Week	Total Marks	50
Submission Date	06-11-2026	Evaluation Date	18-11-2026

MARK SHEET (Rubric-based evaluation)

Task	Title	Max	Awarded	Evidence / evaluation focus
1	Accessing Cloud Services through Different Methods	6		Access-method diagram; demonstration; screenshots; comparison
2	Cloud Platforms and Web Applications	7		Two platforms; interaction diagram; app demonstration; evaluation
3	Web APIs for Cloud Access	7		API design; executable client; two requests; responses
4	Cloud Security and Network Access	7		Secure-access architecture; threats; authentication; mitigations
5	Collaboration Using Cloud-Based Services	6		Shared workspace; permissions; edits; contribution evidence
6	Integrated Cloud Access and Collaboration Mini Investigation	7		Scenario; architecture; prototype; two user tests; evaluation
7	Day-4 Design Challenge: Cloud-Based Collaborative Solution	10		Two independent proposals; comparison; joint design; validation; defence
	TOTAL	**50**		**Percentage:**

OVERALL REMARKS

_____ **Student Signature**	_____ **Faculty Signature**	_____ **Program Director**

MANDATORY SUBMISSION EVIDENCE CHECK

✓ = Present in documented answer; P = Partly present; ✗ = Missing / not satisfied

Task	Evidence expected	Status	Evaluator note
1	Three access methods, client–network–cloud diagram, demonstration and observations		Access-method classification and practical access screenshots to be checked.
2	Two-platform comparison, interaction diagram and cloud web-app demonstration		Platform mapping, user-action sequence and local-versus-cloud evaluation to be checked.
3	API flow, program, two request inputs, responses and execution evidence		Code, actual request/response outputs and analysis to be checked.
4	Secure-access flow, four security considerations, threat and mitigation analysis		Authentication, authorization and secure-communication evidence to be checked.
5	Shared workspace, permission settings, two collaborative activities and contributions		Verify shared editing, user roles, permissions and contribution screenshots.
6	Integrated scenario, architecture, prototype, two user scenarios and results		Verify cloud access, collaboration workflow, tests and limitations.
7	Two initial proposals, comparison record, joint design, tests and cross-defence		Verify independent proposals, joint decisions, observed results and both students' understanding.

IMPLEMENTATION EVIDENCE CHAIN

Evidence item	Status	Observation
Cloud access methods		Browser, mobile and API access methods explained.
Cloud platforms		Two cloud platforms and their applications compared.
Cloud access demonstration		Actual screenshots need verification.
Web API implementation		Program and execution outputs need verification.
Cloud security		Authentication, authorization and security measures explained.
Cloud collaboration		Shared workspace and editing evidence need verification.
Mini-project		Design included; practical testing needs verification.
Day-4 design challenge		Individual proposals, joint design and validation need verification.
Student contributions		Contributions of both students need confirmation.
Screenshots		Original execution screenshots required.

SUBMISSION CHECKLIST (Not Separately Marked)

Requirement	Status
All 7 questions answered	☐ Yes ☐ Partly ☐ No
Cloud access methods explained	☐ Yes ☐ Partly ☐ No

Cloud architecture diagrams included	☐ Yes ☐ Partly ☐ No
Two cloud platforms compared	☐ Yes ☐ Partly ☐ No
Web API program and testing completed	☐ Yes ☐ Partly ☐ No
Cloud security explained	☐ Yes ☐ Partly ☐ No
Cloud collaboration demonstrated	☐ Yes ☐ Partly ☐ No
Mini-project completed and tested	☐ Yes ☐ Partly ☐ No
Day-4 design challenge completed	☐ Yes ☐ Partly ☐ No
Student A and B contributions recorded	☐ Yes ☐ Partly ☐ No
Screenshots and practical evidence attached	☐ Yes ☐ Partly ☐ No
Viva recommended	☐ Yes ☐ No

CLOUD COMPUTING AND BIG DATA ANALYTICS
ASSIGNMENT 3 – CO3

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.*

Mode: Pair Work – 2 Students per Group
Duration: 1 Week **Total Marks:** 50

OBE-Based Evidence Principle

The assignment requires students to demonstrate measurable evidence of:
Understand → Design → Implement → Demonstrate → Analyse → Evaluate
Both students must understand the **complete assignment**. Students should not simply divide questions between them. Individual responsibilities are assigned within each question so that the contribution and attainment of CO3 can be independently verified.

Q1 – Accessing Cloud Services through Different Methods
6 Marks
Shared Group Work
Together, the pair must:

- Identify different methods of accessing cloud services.
- Study browser-based/cloud console access.
- Study access through mobile applications.
- Study access through APIs.
- Identify the role of Internet/network connectivity.
- Select one suitable cloud service for demonstration.

Student A – Access Method Design
Student A must individually:

- Identify and classify at least **three cloud access methods**.
- Prepare a simple architecture/flow diagram showing:

Client → Network → Cloud Service

- Explain how a user accesses the selected cloud service.
- Identify the hardware/software requirements.
- Explain the role of the client and network.
- Justify which access method is most suitable for the selected application.

Student B – Practical Evidence
Student B must individually:

- Demonstrate access to a cloud service using one selected method.
- Record the steps/screenshots as evidence.
- Test the access using different devices or access methods where possible.
- Record observations such as accessibility, response, and usability.
- Identify one limitation encountered during access.
- Explain how the limitation can affect cloud usage.

Cross-Verification
Student A must be able to explain Student B's practical demonstration.
Student B must be able to explain Student A's access architecture.
Evidence Expected: Architecture diagram + screenshots/observations + comparison.

Answer:

1.1 Introduction

Cloud services are delivered over a network and can be reached in several ways. The method chosen decides how fast, how secure and how convenient access is. This answer classifies the access methods, designs the Client → Network → Cloud flow, demonstrates real access to one cloud service (Google Drive) and evaluates the observations.

1.2 Shared Group Work

Both students jointly identified the access methods (browser/console, mobile app, API), studied the role of Internet connectivity, and selected **Google Drive** as the demonstration service because it supports all three access methods and is freely available.

1.3 Classification of Cloud Access Methods (Student A)

Method	How it works	Example with Google Drive	Typical user
Browser / Cloud console	User opens a URL and logs in through a web interface	drive.google.com	General users, administrators
Mobile application	Dedicated app on Android/iOS talks to the cloud service	Google Drive app	Users on the move
API / SDK / CLI	Programs send HTTP requests to service endpoints	Google Drive REST API	Developers, automated systems

1.4 Selected Cloud Service

Google Drive (Storage-as-a-Service, a SaaS offering) was chosen. It stores files on Google's data centres, allows access from any device and provides sharing, versioning and a public API.

1.5 Architecture / Flow: Client → Network → Cloud Service (Student A)

User device (client) → Internet / Wi-Fi / mobile data (network) → Load balancer and authentication layer → Cloud storage service → Response returned to client. The diagram is shown as **Figure 1** at the end of this answer.

1.6 How a User Accesses the Service

- The user opens a browser or app on a device.
- The client sends an HTTPS request to the service domain (DNS resolves it to a Google server).
- The user signs in; credentials are verified (optionally with 2-step verification).
- The cloud service checks the user's permissions and fetches the requested files.
- The response is returned and rendered on the screen.

1.7 Hardware and Software Requirements

Category	Requirement
Hardware	Laptop, desktop, tablet or smartphone; minimum 2 GB RAM; network adapter
Software	Modern browser (Chrome/Edge/Firefox) or the Drive mobile app; OS such as Windows, Android or iOS
Connectivity	Broadband/Wi-Fi/4G or 5G; at least 1–2 Mbps for comfortable use
Account	A valid Google account

1.8 Role of the Client and the Network

The **client** presents the interface, collects user input, encrypts requests (TLS) and displays results. It does not store the master copy of the data. The **network** carries requests and responses; its bandwidth, latency and reliability directly control the user experience, and it is the main link that must be secured (HTTPS).

1.9 Most Suitable Access Method (Justification)

For a student or office user storing and sharing documents, **browser access** is the most suitable: nothing needs installing, it works on any OS, and the full feature set is available. Mobile apps suit on-the-go viewing and uploads, while the API is best for automation. Hence browser access is the primary method, with mobile as secondary.

1.10 Practical Demonstration (Student B)

- Open a browser and visit drive.google.com. **[Insert screenshot]**
- Sign in with a Google account. **[Insert screenshot]**
- Upload a file (for example *Assignment3.pdf*). **[Insert screenshot]**
- Open the same file from the Drive app on a phone. **[Insert screenshot]**
- Confirm that both devices show the same file.

1.11 Observations and Limitation

Device / Method	Login time	Upload response	Usability
Laptop browser	Fast	Quick on Wi-Fi	Full features, very easy
Mobile app	Fast	Slightly slower on mobile data	Easy, smaller screen

Limitation encountered: with a weak or interrupted connection, uploads stalled and files could not be opened. This shows that cloud usage depends on connectivity; users may lose access to work during outages unless offline mode is enabled in advance.

1.12 Cross-Verification

Student A can explain Student B's demonstration: the same file was reached through two clients and one cloud back end. Student B can explain Student A's architecture: each box in the Client → Network → Cloud flow corresponds to a step performed during the demonstration.

1.13 Conclusion

Cloud services can be accessed through browsers, mobile apps and APIs, all following the Client → Network → Cloud pattern. Browser access is the most practical for general users, and the demonstration confirmed device independence as well as the key limitation of network dependence.

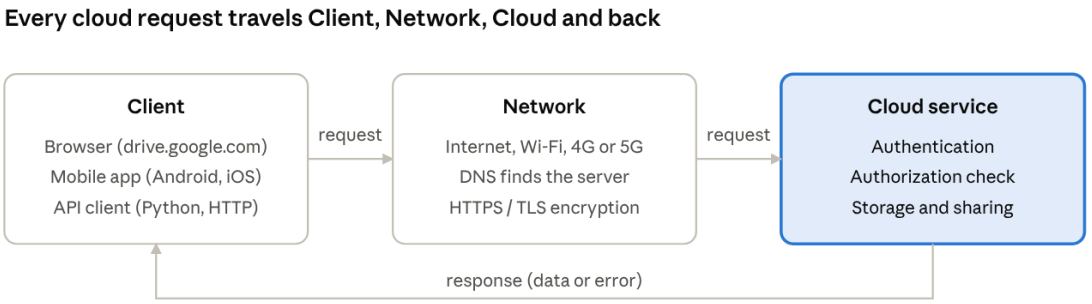


Figure 1 · Client, Network, Cloud flow · 3 stages

Figure 1: the request goes out through the network to the cloud service, and the response returns the same way.

Q2 – Cloud Platforms and Web Applications
7 Marks

Shared Group Work

Together:

- Select **two cloud platforms/services**.
- Identify the web applications available through them.
- Study how users interact with the applications.
- Identify the cloud resources/services supporting the application.
- Prepare a comparative observation.

Student A – Platform Analysis

Student A must individually:

- Identify the major components of the selected cloud platform.
- Explain how the platform supports web applications.
- Map the user request to the cloud service.
- Explain the role of the Internet, server/cloud infrastructure and application.
- Prepare a simple interaction diagram.

Student B – Application Investigation

Student B must individually:

- Select one real-world cloud-based web application.
- Demonstrate its major functions.
- Record the sequence of user actions.
- Identify where cloud access is involved.
- Collect screenshots/evidence of the demonstration.
- Analyse the benefits of using the cloud-based application.

Joint Evaluation

Together answer:

If the same application were installed and operated only on a local computer, what limitations might users experience compared with accessing it through the cloud?

Evidence Expected: Platform/application mapping + demonstration evidence + evaluation.

Answer:

2.1 Introduction

A cloud platform provides computing, storage and networking on which web applications run. This answer compares two platforms, maps a user request to the supporting services, demonstrates a cloud web application and evaluates it against a purely local installation.

2.2 Shared Group Work: Selected Platforms

The pair selected **Amazon Web Services (AWS)** and **Google Cloud Platform (GCP)**, along with their well-known applications (for example AWS WorkDocs/web hosting and Google Workspace/Google Docs).

2.3 Major Components of the Platforms (Student A)

Component	AWS	Google Cloud
Compute	EC2, Lambda	Compute Engine, Cloud Functions
Storage	S3, EBS	Cloud Storage, Persistent Disk
Database	RDS, DynamoDB	Cloud SQL, Firestore
Networking	VPC, Route 53, CloudFront	VPC, Cloud DNS, Cloud CDN
Identity	IAM, Cognito	Cloud IAM, Identity Platform
App platform	Elastic Beanstalk	App Engine

2.4 How the Platform Supports Web Applications

The platform supplies virtual servers or serverless runtimes for the application code, managed databases for data, object storage for files, load balancers for traffic distribution, and auto-scaling to add resources under load. Developers focus on the application while the provider manages hardware, patching and availability.

2.5 Mapping a User Request to Cloud Services (Student A)

Step	User action	Cloud service involved
1	Types the URL	DNS (Route 53 / Cloud DNS)
2	Request reaches the platform	Load balancer / CDN
3	Application logic runs	EC2 / App Engine
4	Data is fetched or saved	Database + object storage
5	Response returned	Application → Internet → browser

2.6 Roles of the Internet, Cloud Infrastructure and Application

The **Internet** is the transport path. The **cloud infrastructure** provides scalable servers, storage and networking. The **application** implements business logic and the user interface. Together they let the user work through a browser without owning any server.

2.7 Interaction Diagram

The interaction sequence between User, Browser, Internet, Load Balancer, Application Server and Database/Storage is shown as **Figure 2** at the end of this answer.

2.8 Application Investigation: Google Docs (Student B)

Google Docs was selected as a real-world cloud web application. Major functions demonstrated: create a document, type and format text, share with another user, comment, and view version history. **[Insert screenshot of each function]**

2.9 Sequence of User Actions and Where Cloud Access Occurs

Sequence	Action	Cloud involvement
1	Open docs.google.com and log in	Authentication service
2	Click New Document	Server creates a file in cloud storage
3	Type text	Autosave sends changes to the cloud every few seconds
4	Click Share	Permission service updates the access list
5	Open on another device	Same file fetched from cloud storage

Benefits observed: no installation, automatic saving, access from anywhere, real-time sharing, automatic backup and free updates.

2.10 Joint Evaluation: Local Installation versus Cloud Access

If the same application ran only on a local computer, users would face these limitations:

- Files could be reached only from that one machine.
- No real-time collaboration; copies would be emailed around.
- Data loss if the disk fails, unless manual backups are made.
- Manual installation, updates and licence management.
- Limited by local CPU, RAM and storage.
- Hard to scale when more users join.

The cloud removes these limits, at the price of needing Internet access and trusting the provider with the data.

2.11 Conclusion

AWS and GCP offer comparable building blocks for hosting web applications. Mapping each user action to a cloud service shows how requests travel from browser to storage. Cloud web applications

such as Google Docs offer accessibility, collaboration and reliability that local installations cannot match.

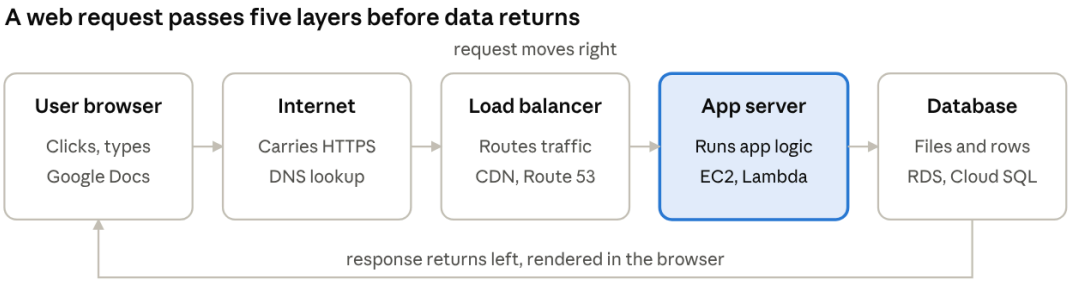


Figure 2 · User request interaction · 5 layers

Figure 2: the application server is the layer that turns a click into data from storage.

Q3 – Web APIs for Cloud Access
7 Marks
Shared Group Work
Together:

- Study the concept of a Web API.
- Identify how APIs enable application-to-cloud communication.
- Select a simple publicly accessible API or cloud API.
- Develop a small program/client to send a request.
- Observe and record the response.

Student A – API Design and Understanding
Student A must individually:

- Explain the purpose of the selected API.
- Identify the request and response components.
- Represent the API communication using a flow diagram.
- Explain the role of:
    - Client
    - API request
    - Cloud/server
    - API response
- Explain how APIs enable programmatic cloud access.

Student B – Implementation and Evidence
Student B must individually:

- Implement a simple API request using Python or another suitable language.
- Execute the program.
- Record input/request and output/response.
- Test at least **two different requests or inputs**.
- Capture execution evidence.
- Analyse the observed results.

Joint Analysis
Together determine:
Why is API-based access useful when a large number of users/applications need to communicate with a cloud service automatically?
Evidence Expected: Program + execution screenshots + request/response evidence + analysis.

Answer:

3.1 Introduction

A Web API lets one program talk to another over HTTP without a human interface. Cloud providers expose most of their features through APIs, which makes automation and integration possible. This answer designs, implements and tests a simple API client.

3.2 Concept of a Web API

A Web API is a set of endpoints (URLs) and rules. A client sends an HTTP request (GET, POST, PUT, DELETE) and the server returns a structured response, usually JSON, along with a status code. REST is the most common style.

3.3 How APIs Enable Application-to-Cloud Communication

Applications embed API calls so they can store files, read data or start services in the cloud without any manual action. Authentication keys or tokens identify the calling application, and standard formats (HTTP + JSON) make the interaction language-independent.

3.4 Selected API and Its Purpose (Student A)

The pair selected the public **JSONPlaceholder** REST API (https://jsonplaceholder.typicode.com), a free, cloud-hosted service that returns sample posts and users. It requires no key, so it is ideal for learning request/response behaviour exactly as a cloud API would behave.

3.5 Request and Response Components (Student A)

Component	Request	Response
Method	GET	–
URL / endpoint	/posts/1	–
Headers	Accept: application/json	Content-Type: application/json
Body	None for GET	JSON data
Status	–	200 OK, 404 Not Found, etc.

3.6 API Communication Flow and Roles

The flow is shown as **Figure 3** at the end of this answer. Roles:

- **Client:** the Python program that builds and sends the request.
- **API request:** an HTTP message that states what is wanted.
- **Cloud/server:** receives the request, processes it and fetches the data.
- **API response:** the status code and JSON data returned to the client.

3.7 Implementation (Student B)

```
import requests

BASE = "https://jsonplaceholder.typicode.com"
```

```
def call_api(endpoint):
    url = BASE + endpoint
    print("REQUEST : GET", url)
    try:
        r = requests.get(url, timeout=10)
        print("STATUS  :", r.status_code)
        print("RESPONSE:", r.json())
    except requests.exceptions.RequestException as e:
        print("ERROR   :", e)
    print("-" * 60)

call_api("/posts/1")   # Request 1
call_api("/users/2")   # Request 2
```

Run with `pip install requests` followed by `python api_client.py`. **[Insert execution screenshot]**

3.8 Two Requests and Their Responses

Request	Input	Status	Response (summary)
1	GET /posts/1	200 OK	JSON with userId 1, id 1 and a title field
2	GET /users/2	200 OK	JSON with name "Ervin Howell", username "Antonette", email and address

Replace with your actual printed output from the screenshot.

3.9 Analysis of Observed Results

Both requests returned status 200 with well-structured JSON in under a second. Changing only the endpoint changed the data returned, showing that one client program can reach many resources. A request to a non-existent endpoint (for example /posts/9999) returns 404, which the program must handle through error checking.

3.10 Joint Analysis: Why API Access Suits Many Users and Applications

API access is useful because it is automatic, fast and standard. Thousands of applications can call the service at the same time without human effort. APIs support scalability, integration across languages and platforms, consistent security (tokens), rate limiting and monitoring. Manual browser access cannot serve such machine-to-machine volume.

3.11 Conclusion

Web APIs are the programmable doorway to cloud services. The Python client successfully sent two requests and received valid responses, confirming how applications communicate with cloud servers automatically and at scale.

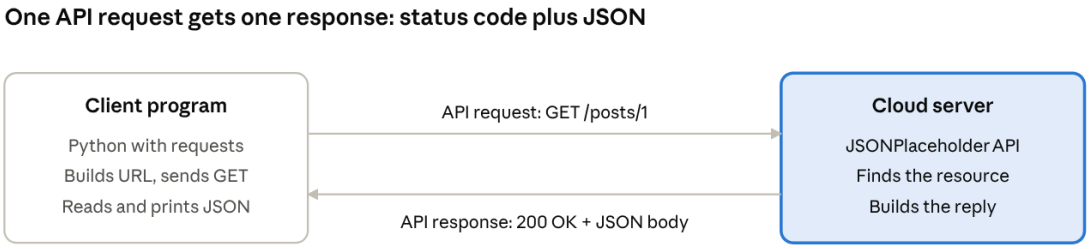


Figure 3 · API request and response · 2 parties

Figure 3: the client sends the request and the cloud server answers with a status code and data.

Q4 – Cloud Security and Network Access
7 Marks
Shared Group Work
Together:

- Identify security issues involved in accessing cloud services.
- Study authentication and authorization.
- Study secure network communication.
- Identify possible threats during cloud access.
- Prepare a secure cloud-access scenario.

Student B – Security Design
Student B must individually:

- Identify at least **four security considerations** for cloud access.
- Explain authentication and authorization.
- Explain the importance of secure communication.
- Design a secure access flow:

User → Authentication → Network → Cloud Service → Authorization → Resource

- Identify possible points of attack.
- Suggest suitable security measures.

Student A – Security Investigation
Student A must individually:

- Investigate a practical cloud-access security scenario.
- Demonstrate or document a suitable authentication mechanism where feasible.
- Compare secure and insecure access situations conceptually.
- Record observations.
- Explain how poor security practices can affect cloud users and data.

Joint Evaluation
Together answer:
If cloud access is available from anywhere through the Internet, what additional security measures are necessary compared with a local application?
Evidence Expected: Secure-access architecture + threat identification + mitigation analysis.

Answer:

4.1 Introduction

Cloud services are reachable from anywhere, so every access passes through networks the user does not control. Security therefore has to protect identity, communication and data. This answer

identifies the security considerations, designs a secure access flow, investigates a practical scenario and lists suitable mitigations.

4.2 Shared Group Work: Security Issues in Cloud Access

The pair identified these issues: stolen or weak passwords, data interception on public Wi-Fi, phishing, excessive permissions, insecure APIs, data leakage through wrong sharing, and loss of data through malware or accidental deletion.

4.3 Four Security Considerations (Student B)

#	Consideration	Why it matters
1	Authentication of users	Only genuine users may enter the service
2	Authorization / least privilege	Users should reach only the resources they need
3	Secure communication (HTTPS/TLS, VPN)	Prevents eavesdropping and tampering on the network
4	Data protection (encryption at rest, backups)	Protects data if storage is breached or lost
5	Monitoring and logging	Detects unusual logins and misuse

4.4 Authentication and Authorization (Student B)

Authentication proves *who you are* (password, OTP, biometric, token). **Authorization** decides *what you may do* after identity is proven (view, edit, delete, administer). A user can be authenticated yet still denied a resource if the permission list does not include them; the system should then return an "access denied" (HTTP 403) message.

4.5 Importance of Secure Communication

Without encryption, passwords and files travel as readable text and can be captured by anyone on the same network. HTTPS uses TLS to encrypt data, verify the server's identity with a certificate and detect tampering. VPNs add a protected tunnel for private access to organisational resources.

4.6 Secure Access Flow (Student B)

The required flow **User → Authentication → Network → Cloud Service → Authorization → Resource** is shown as **Figure 4** at the end of this answer, together with the points of attack and the control placed at each stage.

4.7 Points of Attack and Suitable Security Measures

Stage	Possible attack	Mitigation
User / device	Phishing, malware, keylogger	Awareness training, antivirus, updated OS
Authentication	Password guessing, credential theft	Strong passwords, multi-factor authentication, lockout limits

Stage	Possible attack	Mitigation
Network	Eavesdropping, man-in-the-middle	HTTPS/TLS, VPN, avoid open Wi-Fi
Cloud service	DDoS, API abuse	Firewalls, rate limiting, API keys/tokens
Authorization	Privilege escalation, over-sharing	Role-based access control, least privilege, periodic review
Resource / data	Leakage, ransomware	Encryption, versioning, backups, audit logs

4.8 Security Investigation: Practical Scenario (Student A)

Scenario: a student stores project files on Google Drive and accesses them from a college computer and a phone. Risk: the account is accessed from a shared lab computer on public Wi-Fi.

4.9 Authentication Mechanism Demonstrated (Student A)

Two-step verification was enabled in the Google account: after entering the password, a one-time code or phone prompt is required. **[Insert screenshots: Security settings → 2-Step Verification → login prompt]** The browser address bar also showed the padlock and https://, confirming TLS.

4.10 Secure versus Insecure Access (Conceptual Comparison)

Aspect	Insecure access	Secure access
Protocol	http:// on open Wi-Fi	https:// with certificate
Login	Short, reused password	Strong password + MFA
Sharing	"Anyone with link can edit"	Named users with role-based access
Device	Shared PC, stays logged in	Personal device, log out, session timeout
Result	Data theft, tampering	Controlled and traceable access

4.11 Observations and Effects of Poor Security

With MFA, a correct password alone no longer opened the account. Poor practices can lead to identity theft, deletion or alteration of files, leakage of private data, loss of work, financial damage and loss of trust in the service.

4.12 Joint Evaluation: Extra Measures Compared with a Local Application

A local application is protected mostly by physical access and the local network. A cloud application, reachable from anywhere, additionally needs: strong authentication with MFA, encrypted transport and storage, fine-grained authorization, API protection, continuous monitoring, DDoS protection, regular backups and compliance with data-protection rules.

4.13 Conclusion

Secure cloud access depends on authenticating users, authorizing each action, encrypting communication and protecting data at every stage. Applying layered controls at the identified attack points keeps the advantages of anywhere-access without exposing users or data.

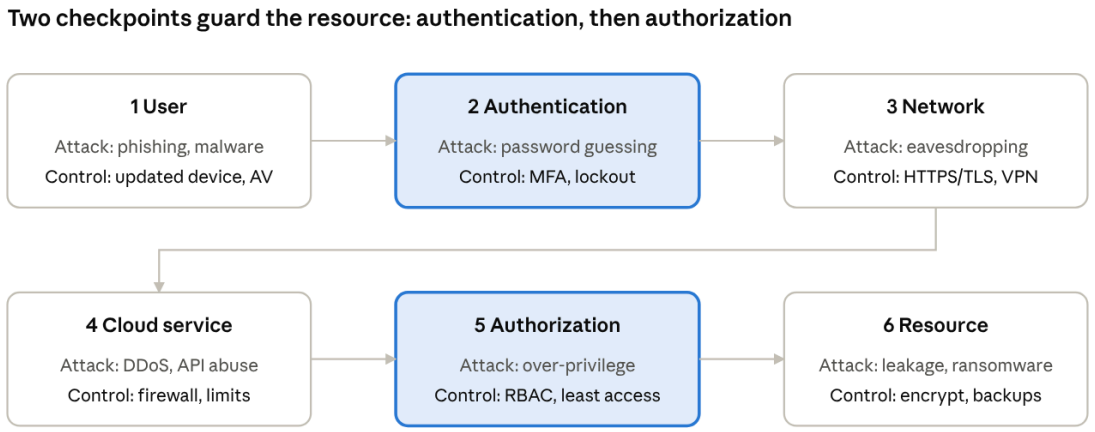


Figure 4 · Secure access flow · 6 stages with attacks and controls

Figure 4: each stage lists its main attack and the control placed against it; the flow continues on the second row.

Q5 – Collaboration Using Cloud-Based Services
6 Marks

Shared Group Work

Together:

- Select a cloud collaboration platform/service.
- Create a shared document/folder/project space.
- Configure appropriate sharing permissions.
- Perform collaborative editing.
- Record the collaboration process.

Student A – Collaboration Design

Student A must individually:

- Identify the collaboration requirements.
- Design a suitable cloud collaboration workflow.
- Explain:
    - Sharing
    - Concurrent access
    - Version control
    - Permission management
- Identify the roles of different users.
- Justify the selected sharing permissions.

Student B – Collaborative Implementation

Student B must individually:

- Participate in the shared cloud workspace.
- Perform at least two collaborative activities.
- Record evidence of contribution.
- Demonstrate sharing/permission settings.

- Observe how changes made by users are reflected.
- Identify one collaboration issue and suggest a solution.

Cross-Understanding
Student B must explain:
How does cloud collaboration reduce the need to exchange multiple copies of the same file?
Student A must explain:
What problems can occur if incorrect sharing permissions are given?
Evidence Expected: Shared workspace + contribution evidence + permission screenshots + observations.

Answer:

5.1 Introduction

Cloud collaboration lets several people work on one shared resource at the same time, instead of passing copies around. This answer designs a collaboration workflow on Google Drive and Google Docs, applies permissions, performs collaborative activities and evaluates the result.

5.2 Shared Group Work: Platform and Workspace

The pair selected **Google Drive / Google Docs**, created a shared folder named *CO3_Assignment_Team* and a shared document *Project_Report*, then configured permissions and edited together. **[Insert screenshot: shared folder and document]**

5.3 Collaboration Requirements (Student A)

- Multiple users edit the same file at once.
- Changes visible to all quickly, with a record of who changed what.
- Ability to restore earlier versions.
- Different access levels for different people.
- Access from any device, with comments for discussion.

5.4 Collaboration Workflow Design (Student A)

- Owner creates the shared folder and document.
- Owner invites members by email with a suitable role.
- Members open the link and edit or comment.
- Changes are saved automatically and merged in real time.
- Version history records every change; the owner can restore any version.
- Final document is reviewed and the permissions are tightened or removed.

5.5 Sharing, Concurrent Access, Version Control and Permission Management

Concept	Explanation	Google Docs feature
Sharing	Giving named users or groups access to one copy	Share button, invite by email
Concurrent access	Several users edit simultaneously without overwriting	Live cursors, real-time merge

Concept	Explanation	Google Docs feature
Version control	Earlier states are stored and restorable	File → Version history
Permission management	Role decides allowed actions	Viewer / Commenter / Editor / Owner

5.6 Roles of Different Users and Justification of Permissions (Student A)

User	Role	Justification
Student A (owner)	Owner	Controls sharing and can delete or transfer the file
Student B	Editor	Must write and modify content
Faculty	Commenter	Gives feedback without changing the content
Classmates (if any)	Viewer	Read-only reference

The principle of least privilege is followed: each person gets only the access needed for the task, and "Anyone with the link" is avoided.

5.7 Permission Settings Demonstrated (Student B)

Share dialog → added Student A as Owner, Student B as Editor and the faculty as Commenter → "General access" kept as *Restricted*. **[Insert permission screenshots]**

5.8 Collaborative Activities Performed (Student B)

Activity	What was done	Evidence
1	Both students edited different sections of the same document at the same time	**[Insert screenshot with two cursors]**
2	Added comments, replied to them and resolved one suggestion	**[Insert comments screenshot]**
3 (optional)	Restored an earlier version from the version history	**[Insert screenshot]**

5.9 Observations: How Changes Are Reflected

Text typed by one student appeared on the other's screen within a few seconds, labelled with the user's cursor colour. Version history listed each edit with name and time, proving individual contribution.

5.10 Collaboration Issue and Solution (Student B)

Issue: two users edited the same paragraph and the text became mixed. **Solution:** assign sections to each person, use *Suggesting* mode for edits to shared text, and communicate through comments.

5.11 Cross-Understanding

How does cloud collaboration reduce multiple copies of a file? (Student B) All users work on one master copy in the cloud, so no email attachments are needed, no "final_v2_new" files exist, and everyone always sees the latest version.

What problems occur with incorrect sharing permissions? (Student A) Editing rights to everyone allow accidental or malicious deletion and tampering; viewer-only rights to a teammate block their work; public links leak confidential data; and ex-members keep access unless removed.

5.12 Conclusion

Cloud collaboration gives one shared copy, real-time editing, version history and role-based permissions. Correct permission design is the key to benefiting from it without risking data or confusion.

Q6 – Integrated Cloud Access and Collaboration Mini Investigation
7 Marks

Students must design a small real-world scenario that requires **both cloud access and cloud collaboration**.

Suggested Scenarios

Choose **one**:

1. Cloud-based Student Project Management
2. Online Student Attendance and Faculty Collaboration
3. Cloud-Based Event Management
4. Cloud-Based Assignment Submission and Evaluation
5. Cloud-Based Hospital Appointment Coordination
6. Cloud-Based Disaster Information Sharing
7. Cloud-Based College Placement Coordination

Student A – System Design

Student A must individually:

- Define the problem.
- Identify users.
- Identify cloud access methods.
- Identify required cloud services.
- Design the system architecture.
- Explain how users access the system.
- Explain how users collaborate.

Student B – Evidence and Evaluation

Student B must individually:

- Develop a small prototype/mock-up/workflow where feasible.
- Demonstrate at least one cloud access mechanism.
- Demonstrate one collaboration mechanism.
- Record screenshots/evidence.
- Test the proposed solution using at least two user scenarios.
- Analyse the results.
- Identify limitations and possible improvements.

Joint Evaluation

Together answer:

How does combining cloud access and cloud collaboration improve the efficiency of the proposed system?

Answer:

6.1 Introduction

This mini investigation combines cloud access and cloud collaboration in one real scenario: **Cloud-Based Assignment Submission and Evaluation**. Students submit work through the cloud and faculty review it collaboratively, replacing paper and email exchange.

6.2 Problem Definition (Student A)

In many colleges, assignments are submitted by email or on paper. Files get lost, late submissions are disputed, versions get mixed up, and faculty cannot give structured feedback quickly. A shared cloud system is needed to submit, track, review and grade work in one place.

6.3 Users Identified (Student A)

User	Needs
Student	Upload assignment, see status, read feedback
Faculty	Download/open submissions, comment, grade
Head of Department / Coordinator	View consolidated results
Administrator	Create folders, manage accounts and permissions

6.4 Cloud Access Methods (Student A)

- **Browser access** to Google Drive, Google Classroom or Forms from any computer.
- **Mobile app** access for students submitting from phones.
- **API access** (optional) for automatically collecting grades into a spreadsheet.

6.5 Required Cloud Services (Student A)

Need	Cloud service
File storage	Google Drive
Submission form with deadline	Google Forms / Classroom
Review and feedback	Google Docs comments, Suggesting mode
Marks register	Google Sheets
Notifications	Gmail / Classroom alerts
Identity	Google account with institutional domain, MFA

6.6 System Architecture (Student A)

Students and faculty → browser/mobile app → Internet (HTTPS) → authentication → Drive/Classroom services → shared folders and marks sheet. The architecture is shown as **Figure 5** at the end of this answer.

6.7 How Users Access and Collaborate (Student A)

Access: users sign in with institutional accounts (plus MFA) through a browser or app; the cloud checks their role before showing folders. **Collaboration:** each student has a private submission folder shared only with faculty; faculty comment directly on submitted documents, and students revise the same file so version history shows the improvement.

6.8 Prototype / Workflow Developed (Student B)

- Created a Drive folder *Assignment_Submissions* with sub-folders per student. **[Insert screenshot]**
- Shared each student folder (Editor) and the faculty (Commenter / Editor). **[Insert screenshot]**
- Created a Google Form as an upload form with a due date. **[Insert screenshot]**
- Created a Sheet *Marks_Register* shared with faculty only. **[Insert screenshot]**

6.9 Mechanisms Demonstrated and Evidence (Student B)

Mechanism	Demonstrated by	Evidence
Cloud access	Logging in from a laptop browser and a phone with the institutional account	**[Insert screenshot]**
Collaboration	Faculty comment on a student's document; student revises it	**[Insert screenshot]**

6.10 Testing with Two User Scenarios (Student B)

Scenario	Action	Expected	Observed
1. Student submits	Student uploads PDF using phone	File appears in own folder, faculty notified	Matches; upload reflected in faculty view *(confirm with your test)*
2. Faculty reviews	Faculty opens file, adds comment and mark	Student sees comment; marks hidden from other students	Matches; other student could not open the folder *(confirm with your test)*

6.11 Analysis of Results, Limitations and Improvements

Both tests behaved as designed: access was controlled by role and feedback was visible quickly. **Limitations:** depends on Internet connectivity, free storage limits, manual folder creation, and no plagiarism check. **Improvements:** automate folder creation with Apps Script/API, add deadline reminders, integrate plagiarism tools and enable offline access.

6.12 Joint Evaluation: Why Combining Access and Collaboration Improves Efficiency

Access from any device removes waiting for physical presence; collaboration on one shared file removes copying and reconciliation. Together they give faster submission, instant feedback, one source of truth, a time-stamped record and fewer lost files.

6.13 Conclusion

The assignment submission system shows that cloud access plus collaboration saves time, improves transparency and makes evaluation traceable, while role-based permissions keep student data protected.

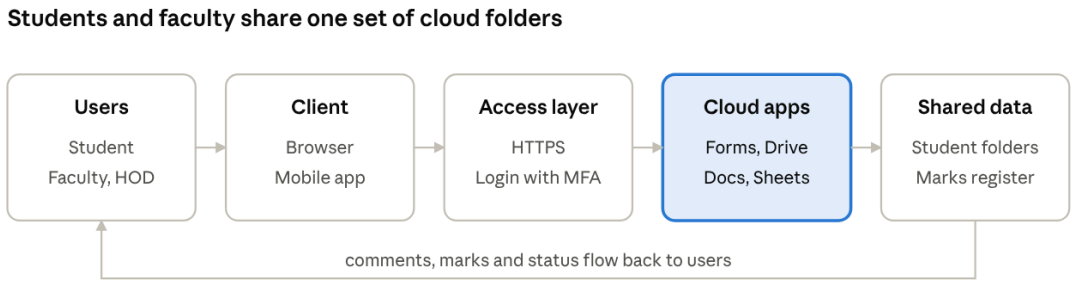


Figure 5 · Assignment submission architecture · 5 layers

Figure 5: access is controlled at the access layer, while the cloud apps provide collaboration on the shared data.

Q7 – DAY-4 DESIGN CHALLENGE

Design a Cloud-Based Collaborative Solution
10 Marks

This question must be solved **collaboratively**.

Neither student should independently prepare the complete solution and combine it later.

Stage 1 – Independent Design Proposal
Before discussing with each other:
Student A independently proposes:

- Problem definition
- Target users
- Cloud access method
- Required cloud services
- Collaboration mechanism
- Basic architecture

Student B independently proposes:

- Problem definition
- Target users
- Cloud access method
- Required cloud services
- Collaboration mechanism
- Basic architecture

Both initial proposals must be retained as evidence.

Stage 2 – Compare the Two Designs

Together discuss:

- Where the designs agree
- Where they differ
- Strengths and weaknesses
- Security concerns
- Accessibility
- Collaboration requirements
- Missing components
- Which ideas should be retained

Prepare a **Design Comparison Record**.

Stage 3 – Joint Final Design

Develop the final cloud-based solution.

The final design must identify:

- Problem
- Users
- Cloud access method
- Network requirements
- Cloud service/platform
- Web application/API requirement
- Collaboration mechanism
- Authentication/security
- Data/resource sharing
- Complete system workflow

Prepare an architecture such as:

Users → Client → Internet/Network → Cloud Platform → Application/API → Shared Resources

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Stage 4 – Validation

Student A – Technical Validation

Student A must:

- Explain the selected cloud access method.
- Justify the platform/service selection.
- Explain the system architecture.
- Explain how API/web access is used, if applicable.
- Analyse the suitability of the design.

Student B – Experimental Validation

Student B must:

- Design test cases.
- Demonstrate the solution/workflow.
- Test different users/access conditions.
- Record observations.
- Present screenshots or other evidence.
- Compare expected and observed behaviour.

Stage 5 – Cross-Defence

Student A must explain:
What experimental evidence supports the group's cloud design?

Student B must explain:
Why was the selected cloud access and collaboration approach chosen?

Stage 6 – Joint Evaluation

Together determine:

- Benefits of the proposed cloud solution
- Access-related limitations
- Collaboration benefits
- Security risks
- Scalability considerations
- Possible improvements
- Situations where the proposed approach may not be appropriate

Final Question

What characteristics should a system designer consider before selecting cloud-based access and collaboration for a real-world application?

Answer:

7.1 Introduction

This challenge follows six stages: independent proposals, comparison, joint design, validation, cross-defence and joint evaluation. The common theme chosen is campus event coordination, where many users must reach and edit shared information from different places.

7.2 Stage 1A: Student A's Independent Proposal (retained as evidence)

Element	Proposal
Problem	Placement drive details are shared by WhatsApp and email, causing confusion
Target users	Placement officer, students, company HR
Access method	Mobile app and browser
Cloud services	Google Drive, Sheets, Forms
Collaboration	Shared Sheet for schedules and a shared folder for resumes
Basic architecture	Users → Mobile/Browser → Internet → Google Workspace → Shared Sheet/Folder

7.3 Stage 1B: Student B's Independent Proposal (retained as evidence)

Element	Proposal
Problem	College events are planned on paper; tasks, budgets and registrations are not tracked
Target users	Event coordinators, volunteers, faculty, participants
Access method	Browser web application with a REST API for registration data
Cloud services	Cloud-hosted web app, database, object storage
Collaboration	Shared task board and shared document for event plans
Basic architecture	Users → Browser → Internet → Web App/API → Database and Storage

7.4 Stage 2: Design Comparison Record

Aspect	Agreement	Difference	Decision
Problem	Both address coordination by many people	Placement versus events	Choose event coordination (broader users, easier testing)
Access	Both use browser	A: mobile focus; B: API	Browser + mobile + small API
Services	Both use cloud storage	A: ready SaaS; B: custom app	SaaS tools plus a light API for registration
Collaboration	Shared documents	A: sheets; B: task board	Shared sheet/board with roles
Security	A: basic sharing; B: not stated	Both missed MFA	Add MFA and role-based access

Strengths and weaknesses: A's design is simple and quick but limited to one function; B's is flexible but needs more development. **Missing in both:** authentication detail, backup plan, offline behaviour. **Retained:** A's mobile access and shared sheet, B's API and task board.

7.5 Stage 3: Joint Final Design (Problem, Users, Access, Network)

- **Problem:** manual and scattered coordination of college events.
- **Users:** organiser (owner), team members (editors), faculty (commenters), participants (form users).
- **Access method:** browser and mobile app; API for registration data.
- **Network requirements:** reliable Wi-Fi/4G, HTTPS only, at least 2 Mbps per user.

7.6 Stage 3: Joint Final Design (Platform, API, Collaboration, Security, Sharing)

Item	Decision
Cloud platform	Google Workspace for storage and collaboration, with a small cloud-hosted API for registrations
Web application / API	Google Forms for registration; REST call returns participant counts
Collaboration	Shared Sheet (task list and budget), shared Doc (event plan), comments
Authentication / security	Institutional login with MFA, HTTPS, role-based roles, restricted link sharing
Data sharing	Folder per event; participant data visible only to organisers

7.7 Stage 3: Complete System Workflow and Architecture

- Organiser creates the event folder, sheet and registration form.
- Team members are invited with role-based permissions.
- Participants register through a form on browser or phone.
- Responses are stored in the shared sheet; the API returns counts.
- Team updates tasks and budget concurrently; faculty comment.
- Final report is produced and archived.

The architecture **Users → Client → Internet/Network → Cloud Platform → Application/API → Shared Resources** is shown as **Figure 6** at the end of this answer.

7.8 Stage 4A: Technical Validation (Student A)

Browser/mobile access was chosen because organisers and participants use varied devices without installation. Google Workspace was selected for its built-in sharing, version history and security, avoiding the cost of building a platform. The API gives programmatic participant counts. The design suits the need because it is low-cost, scalable to hundreds of users and easy to learn; its weakness is Internet dependency.

7.9 Stage 4B: Test Cases (Student B)

ID	Test	Expected result
T1	Participant registers from phone	Row appears in sheet within seconds
T2	Two organisers edit the task sheet together	Both edits retained
T3	Volunteer (Viewer) tries to edit budget	Edit blocked
T4	Unauthenticated user opens the folder link	Access denied
T5	Poor network during registration	Form retries or shows error, no duplicate data

ID	Test	Expected result
T6	API call for participant count	Status 200 with correct count

7.10 Stage 4B: Demonstration, Observations and Comparison

ID	Observed behaviour	Match?
T1	*(record your result, e.g. row appeared in about 3 s)*	☑ / ☐
T2	*(record)*	☑ / ☐
T3	*(record: "You need edit access")*	☑ / ☐
T4	*(record: sign-in/denied page)*	☑ / ☐
T5	*(record)*	☑ / ☐
T6	*(record: 200 and count)*	☑ / ☐

[Insert screenshots for each test.] Fill the table with your actual results; any mismatch should be explained as a design lesson.

7.11 Stage 5: Cross-Defence

Student A – What experimental evidence supports the design? T1 and T2 show access and real-time collaboration work from different devices; T3 and T4 show authorization and authentication are enforced; T6 proves API access works. Together these satisfy the access, collaboration and security requirements.

Student B – Why was this access and collaboration approach chosen? It needs no installation, works on every device, supports simultaneous editing with version history, allows role-based permission, and is cheaper and faster to deploy than a custom system.

7.12 Stage 6: Joint Evaluation

Criterion	Evaluation
Benefits	Single source of truth, real-time updates, low cost, quick setup
Access limitations	Needs Internet and a compatible device; accounts required
Collaboration benefits	Concurrent editing, comments, history, fewer copies
Security risks	Account compromise, wrong sharing, data leakage
Scalability	Handles hundreds of users; very large events need a dedicated database and API gateway
Possible improvements	Automated notifications, dashboard, offline mode, audit logging
Not appropriate when	Connectivity is unreliable, data is highly confidential or regulated, or strict data-residency rules apply

7.13 Final Question: Characteristics a System Designer Should Consider

Before choosing cloud access and collaboration, a designer should consider: user count and location; connectivity reliability; device variety; sensitivity and regulation of data; security and authentication needs; required collaboration level; scalability and performance; cost model; availability and backup; integration through APIs; vendor lock-in and exit options; and user training and usability.

7.14 Conclusion

Starting from two independent proposals and merging their strengths produced a more complete design than either alone. The tests confirm that browser/mobile access, shared workspaces, role-based permissions and a small API together form a practical collaborative cloud solution, with clearly understood limits.

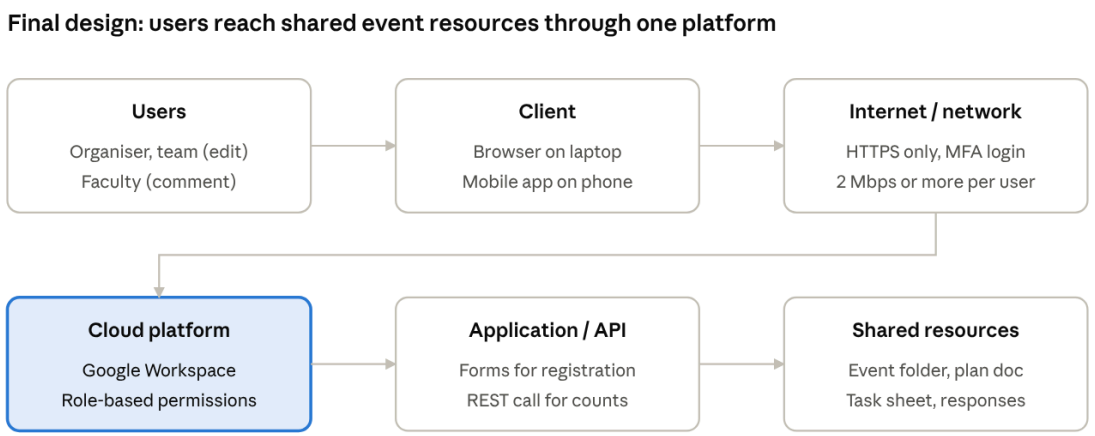


Figure 6 · Final cloud design architecture · 6 stages

Figure 6: the flow runs left to right, then continues on the second row into the shared resources.

CONTRIBUTION RECORD

Each pair must include the following table at the end of the submission.

Question	Student A – Primary Contribution	Student B – Primary Contribution	Joint Work
Q1	Access-method design and architecture	Practical access demonstration and screenshots	Service selection, cross-verification
Q2	Platform analysis and interaction diagram	Application investigation and screenshots	Platform selection, local-vs-cloud evaluation

Q3	API design, flow diagram, explanation	Python implementation and execution	API selection, joint analysis
Q4	Security investigation and MFA demonstration	Security design and attack/mitigation analysis	Scenario preparation, joint evaluation
Q5	Collaboration design and roles	Collaborative activities and evidence	Workspace creation, cross-understanding
Q6	System design and architecture	Prototype, tests and results	Scenario choice, efficiency evaluation
Q7	Technical validation, own proposal	Test cases, demonstration, own proposal	Comparison, joint design, evaluation

The contribution record must reflect the **actual work completed**.
Primary contribution does not mean exclusive responsibility. Both students are accountable for the complete assignment.

INDIVIDUAL VERIFICATION – OBE EVIDENCE
Faculty may select any question or component for individual verification.
Each student must be able to:

- Explain the cloud access method.
- Explain the role of client, network and cloud service.
- Explain web application/API access.
- Explain security considerations.
- Demonstrate cloud collaboration.
- Interpret experimental evidence.
- Justify design decisions.
- Respond to a modified requirement or scenario.

A student should not answer:
“My partner did this section.”
Understanding of the complete submitted work is mandatory.

RECOMMENDED FACULTY VERIFICATION MODEL – VIVA
For each student, conduct a brief **4–5 minute verification**.
Question 1 – Primary Responsibility
Ask one technical question from the component for which the student had primary responsibility.
Question 2 – Partner's Responsibility
Ask one question from a component primarily completed by the partner.
Question 3 – Transfer Question
Modify a condition and ask the student to reason about the consequence.
Examples:
“If Internet connectivity becomes unreliable, how would it affect cloud access?”

“If a shared cloud document is given editing permission to everyone, what problems could occur?”
“If the API becomes unavailable, how would your application behave?”
“If a user is authenticated but does not have permission to access a resource, what should happen?”
This provides **individual evidence of CO3 attainment** without requiring two completely separate assignments.

OBE Evidence Mapping

CO3 Component	**Assignment Evidence**
Build methods to access cloud	Q1, Q2, Q3
Access through platforms/web applications	Q2
Programmatic/API-based access	Q3
Secure cloud access	Q4
Collaboration within cloud	Q5
Apply cloud access + collaboration	Q6
Design and evaluate a cloud solution	Q7
Individual attainment	Viva + contribution record
Practical evidence	Screenshots, programs, observations, test results
Higher-order learning	Design, analysis, evaluation and transfer questions